

LETICIA PIERAERTS

ECARES, Université libre de Bruxelles Avenue F.D Roosevelt 42, 1050 Brussels, Belgium

✉ leticia.pieraerts@ulb.be 🌐 [Personal website](#)

Belgian and Spanish. Born April 9, 1997.

RESEARCH INTERESTS

Environmental and Energy Economics, Electricity markets, Theoretical and Empirical Industrial Organization

POSITION

Université libre de Bruxelles, Belgium

Oct 2021 – now

PhD Candidate in Economics

Supervisor: Estelle Cantillon

Grant from the Federal Energy Transition Fund ([Project Demandflex](#) [🔗](#))

EDUCATION

Toulouse School of Economics

Jan 2026 – April 2026

Visiting PhD Candidate in Environmental research group

Host: Stefan Ambec

Université libre de Bruxelles, Belgium

Sept 2021 – Sept 2023

MRes Research in Economics

Magna cum Laude

Université libre de Bruxelles, Belgium

Sept 2019 – Aug 2021

MSc Economics

Exchange semester at the Technical University of Munich (Germany)

Quantitative Techniques for Economics and Management (QTEM) network

Magna cum Laude

Université libre de Bruxelles, Belgium

Sept 2015 – Aug 2018

BSc Economics

Magna cum Laude

RESEARCH PROJECTS

Vertical arrangements and market power with real-time pricing: Evidence from the Spanish electricity market

Abstract: In electricity markets, vertical integration between production and retail commitments has been shown to be pro-competitive: it reduces producers' net position on the spot market, lowering their incentive to exercise market power (Bushnell et al., 2008). This result holds under fixed-price contracts, where consumers face a fixed price for the entire delivery period. With real-time pricing, retailers no longer need to commit to a retail price before participating on the spot market. How does this affect price formation on the retail and spot market? I develop a model of strategic behavior of vertically integrated firms with real-time pricing and find that the pro-competitive effect of vertical integration disappears entirely. I test the model predictions using microdata from the Spanish electricity market where real-time pricing was introduced as the default retail contract in 2015. I find preliminary evidence that the observed changes in spot prices and firm behavior are consistent with the model.

Electricity market architecture, contracts and real-time price exposure *with Estelle Cantillon*

Abstract: Demand-side flexibility is essential to the energy transition. Despite accounting for a significant share of total electricity consumption, industrial consumers exhibit limited demand-response. This paper looks at the role of contractual arrangements in mediating industrial demand-response. Leveraging a unique dataset of retail contracts for large industrial consumers in Belgium over 2014-2024, we document substantial heterogeneity in contract types and actual exposure to the spot market. Preliminary results suggest a small but significant demand-response despite low spot exposure. To account for endogeneity of contract choice, we construct a shift-share instrument that combines the availability of flexible technologies by sector with firm-level takeup within sectors. These findings have direct implications for policies aimed at unlocking industrial demand-response.

Who benefits from the energy transition? Profits and welfare in retail electricity markets *with Elise Viadere*

Abstract: The energy transition is reshaping retail electricity markets through rising wholesale price volatility from intermittent renewables, growing consumer flexibility, and regulation requiring retailers to offer both fixed-price and dynamic contracts. We study how these forces shape a monopolist retailer's pricing strategy, profits and welfare. We characterize optimal contracts across three portfolio structures that correspond to the stages of the transition: fixed-price contracts only (initial), a menu of fixed-price and dynamic contracts (transition), and dynamic contracts only (final). In the initial and transition stages, the retailer extracts all consumer surplus and raises profits through price discrimination. In the final stage, dynamic prices are constrained by low-flexibility consumers' participation constraint, leading to less volatile retail prices than in the wholesale market, but profits still increase compared to menu pricing if the share of flexible consumers is sufficiently low. Consumer surplus only accrues to high-flexibility consumers, who are typically higher-income, raising concerns about the distributional consequences of the energy transition.

PUBLICATIONS

Cantillon, E., & Pieraerts, L. (2023). La transition énergétique en Belgique : États des lieux. In E. Cantillon, S. Brunet, J. Eyckmans, A. Gautier, M. Muûls, & F. Venmans (Eds.), *Réussir la transition vers une économie zéro carbone*. Editions de l'Université Ouverte.

POLICY WORK

Study on cost pass-through in energy retail markets 2018 – 24 (Belgium, France, Germany, Spain), for the European Commission – DG ENER (with Estelle Cantillon, Stefan Lamp, Mar Reguant, Isabelle Méjean, Mattia Nardotto and Elise Viadere)

SEMINARS, CONFERENCES AND SUMMER SCHOOLS

* = planned

2026: TSE Environmental Workshop (Toulouse), Energy Workshop (Madrid), EDF research seminar (Paris Saclay), ENTER Conference* (Madrid)

2025: Young Energy Economists and Engineers Seminar 2025 (Paris), Climate Policy Summer School (Berlin), FAEE PhD conference (Grenoble), CREG internal seminar (Brussels), ECARES PhD Day (Brussels), AEE annual conference [poster] (Paris)

2024: ECARES internal seminar, ECARES PhD seminar (Brussels)

2023: SRBE Study Day on Flexible Loads (Brussels), ETF Project Conference (Ghent), ECARES PhD seminar (Brussels)

2022: Energy Workshop (TSE), Summer School on Empirical Methods for the Analysis of the Energy Transition (BSE, Barcelona)

TEACHING AND RESEARCH EXPERIENCE

TA Advanced Industrial Organization, Faculty of Economics (2023)

For Mattia Nardotto

TA Energy Policy and Management, Ecole Polytechnique de Bruxelles (2023, 2024, 2025)

For Benjamin Genêt

Research Assistant, ECARES (June-Oct 2021)

For Estelle Cantillon

- Data cleaning and processing on the EU Emissions Trading Scheme

WORK EXPERIENCE

**National Organization for Hydrogen and Fuel Cell Battery (NOW GmbH),
Berlin (Germany)**

Sept 2018 – Aug 2019

Internship in Communication and Knowledge Management

TECHNICAL AND LANGUAGE SKILLS

Programming/statistical software: Python, R, STATA, $\text{\LaTeX}$

Collaboration software: Git, Github, Overleaf

Languages: French (native), Spanish (native), English (C2), German (B2+)